Machine Learning Fundamentals

Machine learning is a subset of artificial intelligence that focuses on building systems that learn from data. Instead of being explicitly programmed, these systems use algorithms to identify patterns and make decisions.

There are three main types of machine learning:
1. Supervised Learning - uses labeled data to train models
2. Unsupervised Learning - finds patterns in unlabeled data
3. Reinforcement Learning - learns through trial and error

Common algorithms include linear regression, decision trees, random forests, support vector machines, and neural networks. Deep learning, a subset of machine learning, uses multi-layered neural networks to model complex patterns in large datasets.

Natural Language Processing

Natural Language Processing (NLP) is a field of AI that enables computers to understand, interpret, and generate human language.

Key NLP tasks include:
- Text classification and sentiment analysis
- Named entity recognition (NER)
- Machine translation
- Question answering
- Text summarization

Modern NLP relies heavily on transformer architectures such as BERT, GPT, and T5. These models are pre-trained on large corpora and can be fine-tuned for specific downstream tasks.

Vector embeddings are central to NLP. They represent words, sentences, or documents as dense numerical vectors that capture semantic meaning. Similar texts have similar vector representations.

Computer Vision

Computer vision is a field of AI that trains computers to interpret and understand visual information from the world, such as images and videos.

Core computer vision tasks include:
- Image classification
- Object detection and localization
- Image segmentation (semantic and instance)
- Facial recognition
- Optical character recognition (OCR)

Convolutional Neural Networks (CNNs) have been the backbone of computer vision for years. Architectures like ResNet, VGG, and EfficientNet achieve high accuracy on image benchmarks.

More recently, Vision Transformers (ViT) have shown that transformer architectures can also excel at visual tasks, often outperforming traditional CNN approaches on large-scale datasets.